```
# Install necessary libraries
!pip install chromadb
!pip install sentence-transformers
```

```
  Downloading pybase64-1.4.3-cp312-cp312-manylinux1_x86_64.manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_5_x86_64.whl
Requirement already satisfied: uvicorn>=0.18.3 in /usr/local/lib/python3.12/dist-packages (from uvicorn[standard]>=0.18.3->chro
Requirement already satisfied: numpy>=1.22.5 in /usr/local/lib/python3.12/dist-packages (from chromadb) (2.0.2)
Requirement already satisfied: typing-extensions>=4.5.0 in /usr/local/lib/python3.12/dist-packages (from chromadb) (4.15.0)
Collecting onnxruntime>=1.14.1 (from chromadb)
  Downloading onnxruntime-1.24.4-cp312-cp312-manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl.metadata (5.2 kB)
Requirement already satisfied: opentelemetry-api>=1.2.0 in /usr/local/lib/python3.12/dist-packages (from chromadb) (1.38.0)
Collecting opentelemetry-exporter-otlp-proto-grpc>=1.2.0 (from chromadb)
  Downloading opentelemetry_exporter_otlp_proto_grpc-1.40.0-py3-none-any.whl.metadata (2.6 kB)
Requirement already satisfied: opentelemetry-sdk>=1.2.0 in /usr/local/lib/python3.12/dist-packages (from chromadb) (1.38.0)
Requirement already satisfied: tokenizers>=0.13.2 in /usr/local/lib/python3.12/dist-packages (from chromadb) (0.22.2)
Collecting pypika>=0.48.9 (from chromadb)
  Downloading pypika-0.51.1-py2.py3-none-any.whl.metadata (51 kB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 52.0/52.0 kB 2.6 MB/s eta 0:00:00
Requirement already satisfied: tqdm>=4.65.0 in /usr/local/lib/python3.12/dist-packages (from chromadb) (4.67.3)
Requirement already satisfied: overrides>=7.3.1 in /usr/local/lib/python3.12/dist-packages (from chromadb) (7.7.0)
Requirement already satisfied: importlib-resources in /usr/local/lib/python3.12/dist-packages (from chromadb) (6.5.2)
Requirement already satisfied: grpcio>=1.58.0 in /usr/local/lib/python3.12/dist-packages (from chromadb) (1.80.0)
Collecting bcrypt>=4.0.1 (from chromadb)
  Downloading bcrypt-5.0.0-cp39-abi3-manylinux_2_34_x86_64.whl.metadata (10 kB)
Requirement already satisfied: typer>=0.9.0 in /usr/local/lib/python3.12/dist-packages (from chromadb) (0.24.1)
Collecting kubernetes>=28.1.0 (from chromadb)
  Downloading kubernetes-35.0.0-py2.py3-none-any.whl.metadata (1.7 kB)
Requirement already satisfied: tenacity>=8.2.3 in /usr/local/lib/python3.12/dist-packages (from chromadb) (9.1.4)
Requirement already satisfied: pyyaml>=6.0.0 in /usr/local/lib/python3.12/dist-packages (from chromadb) (6.0.3)
Requirement already satisfied: mmh3>=4.0.1 in /usr/local/lib/python3.12/dist-packages (from chromadb) (5.2.1)
Requirement already satisfied: orjson>=3.9.12 in /usr/local/lib/python3.12/dist-packages (from chromadb) (3.11.8)
Requirement already satisfied: httpx>=0.27.0 in /usr/local/lib/python3.12/dist-packages (from chromadb) (0.28.1)
Requirement already satisfied: rich>=10.11.0 in /usr/local/lib/python3.12/dist-packages (from chromadb) (13.9.4)
Requirement already satisfied: jsonschema>=4.19.0 in /usr/local/lib/python3.12/dist-packages (from chromadb) (4.26.0)
Requirement already satisfied: packaging>=24.0 in /usr/local/lib/python3.12/dist-packages (from build>=1.0.3->chromadb) (26.0)
Collecting pyproject_hooks (from build>=1.0.3->chromadb)
  Downloading pyproject_hooks-1.2.0-py3-none-any.whl.metadata (1.3 kB)
Requirement already satisfied: anyio in /usr/local/lib/python3.12/dist-packages (from httpx>=0.27.0->chromadb) (4.13.0)
Requirement already satisfied: certifi in /usr/local/lib/python3.12/dist-packages (from httpx>=0.27.0->chromadb) (2026.2.25)
Requirement already satisfied: httpcore==1.* in /usr/local/lib/python3.12/dist-packages (from httpx>=0.27.0->chromadb) (1.0.9)
Requirement already satisfied: idna in /usr/local/lib/python3.12/dist-packages (from httpx>=0.27.0->chromadb) (3.11)
Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.12/dist-packages (from httpcore==1.*->httpx>=0.27.0->chroma
Requirement already satisfied: attrs>=22.2.0 in /usr/local/lib/python3.12/dist-packages (from jsonschema>=4.19.0->chromadb) (2
Requirement already satisfied: jsonschema-specifications>=2023.03.6 in /usr/local/lib/python3.12/dist-packages (from jsonschema
Requirement already satisfied: referencing>=0.28.4 in /usr/local/lib/python3.12/dist-packages (from jsonschema>=4.19.0->chroma
Requirement already satisfied: rpds-py>=0.25.0 in /usr/local/lib/python3.12/dist-packages (from jsonschema>=4.19.0->chromadb)
Requirement already satisfied: six>=1.9.0 in /usr/local/lib/python3.12/dist-packages (from kubernetes>=28.1.0->chromadb) (1.17
Requirement already satisfied: python-dateutil>=2.5.3 in /usr/local/lib/python3.12/dist-packages (from kubernetes>=28.1.0->chro
Requirement already satisfied: websocket-client!=0.40.0,!=0.41.*,!=0.42.*,>=0.32.0 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: requests in /usr/local/lib/python3.12/dist-packages (from kubernetes>=28.1.0->chromadb) (2.32.4
Requirement already satisfied: requests-oauthlib in /usr/local/lib/python3.12/dist-packages (from kubernetes>=28.1.0->chromadb
Requirement already satisfied: urllib3!=2.6.0,>=1.24.2 in /usr/local/lib/python3.12/dist-packages (from kubernetes>=28.1.0->ch
Collecting durationpy>=0.7 (from kubernetes>=28.1.0->chromadb)
  Downloading durationpy-0.10-py3-none-any.whl.metadata (340 bytes)
Requirement already satisfied: flatbuffers in /usr/local/lib/python3.12/dist-packages (from onnxruntime>=1.14.1->chromadb) (25
Requirement already satisfied: protobuf in /usr/local/lib/python3.12/dist-packages (from onnxruntime>=1.14.1->chromadb) (5.29.6
Requirement already satisfied: sympy in /usr/local/lib/python3.12/dist-packages (from onnxruntime>=1.14.1->chromadb) (1.14.0)
Requirement already satisfied: importlib-metadata<8.8.0,>=6.0 in /usr/local/lib/python3.12/dist-packages (from opentelemetry-ap
Requirement already satisfied: googleapis-common-protos~=1.57 in /usr/local/lib/python3.12/dist-packages (from opentelemetry-ex
Collecting opentelemetry-exporter-otlp-proto-common==1.40.0 (from opentelemetry-exporter-otlp-proto-grpc>=1.2.0->chromadb)
  Downloading opentelemetry_exporter_otlp_proto_common-1.40.0-py3-none-any.whl.metadata (1.9 kB)
Collecting opentelemetry-proto==1.40.0 (from opentelemetry-exporter-otlp-proto-grpc>=1.2.0->chromadb)
  Downloading opentelemetry_proto-1.40.0-py3-none-any.whl.metadata (2.4 kB)
```

```
# Import libraries
import chromadb
from sentence_transformers import SentenceTransformer
```

```
client = chromadb.Client()
collection = client.create_collection(name="rag_collection", get_or_create=True)
print(f"ChromaDB collection '{collection.name}' initialized.")
```

```
ChromaDB collection 'rag_collection' initialized.
```

```
model = SentenceTransformer('all-MiniLM-L6-v2')
print("SentenceTransformer model 'all-MiniLM-L6-v2' loaded.")
```

```
/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:94: UserWarning:
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (https://huggingface.co/settings/tokens), set it
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access public models or datasets.
  warnings.warn(
modules.json: 100%  349/349 [00:00<00:00, 36.6kB/s]
config_sentence_transformers.json: 100%  116/116 [00:00<00:00, 12.1kB/s]
README.md: 10.5k/? [00:00<00:00, 889kB/s]
sentence_bert_config.json: 100%  53.0/53.0 [00:00<00:00, 870B/s]
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster d
WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to en
config.json: 100%  612/612 [00:00<00:00, 15.1kB/s]
model.safetensors: 100%  90.9M/90.9M [00:01<00:00, 209MB/s]
Loading weights: 100%  103/103 [00:00<00:00, 379.47it/s, Materializing param=pooler.dense.weight]
BertModel LOAD REPORT from: sentence-transformers/all-MiniLM-L6-v2
Key                     | Status     |  |
------------------------+------------+--+-
embeddings.position_ids | UNEXPECTED |  |

Notes:
- UNEXPECTED    :can be ignored when loading from different task/architecture; not ok if you expect identical arch.
tokenizer_config.json: 100%  350/350 [00:00<00:00, 10.9kB/s]
vocab.txt: 232k/? [00:00<00:00, 5.16MB/s]
tokenizer.json: 466k/? [00:00<00:00, 15.9MB/s]
special_tokens_map.json: 100%  112/112 [00:00<00:00, 1.56kB/s]
config.json: 100%  190/190 [00:00<00:00, 4.59kB/s]
SentenceTransformer model 'all-MiniLM-L6-v2' loaded.
```

```
documents = [
    "Artificial Intelligence is the simulation of human intelligence.",
    "Machine Learning is a subset of AI.",
    "Deep Learning uses neural networks.",
    "Natural Language Processing deals with text data.",
    "RAG combines retrieval and generation."
]

ids = ["doc1", "doc2", "doc3", "doc4", "doc5"]

print("Sample documents prepared.")
```

```
Sample documents prepared.
```

```
embeddings = model.encode(documents)
print(f"Converted {len(documents)} documents into {embeddings.shape[1]}-dimensional embeddings.")
```

```
Converted 5 documents into 384-dimensional embeddings.
```

```
collection.add(
    documents=documents,
    embeddings=embeddings.tolist(), # Convert numpy array to list for ChromaDB
    ids=ids
)
print(f"Added {len(documents)} documents and their embeddings to ChromaDB collection '{collection.name}'.")
```

```
Added 5 documents and their embeddings to ChromaDB collection 'rag_collection'.
```

```
query = "What is AI?"

# Encode the query into an embedding
query_embedding = model.encode([query])

# Perform similarity search in ChromaDB
results = collection.query(
    query_embeddings=query_embedding.tolist(), # Convert numpy array to list
    n_results=2 # Retrieve the top 2 most similar documents
)
```

```
print("Similarity search results:")
print(results)
```

```
Similarity search results:
{'ids': [['doc1', 'doc2']], 'embeddings': None, 'documents': [['Artificial Intelligence is the simulation of human intelligence.
```

```
print("Retrieved Documents:")
for doc in results['documents'][0]:
    print(f"- {doc}")
```

```
Retrieved Documents:
- Artificial Intelligence is the simulation of human intelligence.
- Machine Learning is a subset of AI.
```

```
def retrieve(query):
    query_embedding = model.encode([query])
    results = collection.query(
        query_embeddings=query_embedding.tolist(),
        n_results=2 # You can adjust this number
    )
    return results['documents'][0]

print("Testing retrieval function for 'Explain machine learning':")
retrieved_ml_docs = retrieve("Explain machine learning")
for doc in retrieved_ml_docs:
    print(f"- {doc}")
```

```
Testing retrieval function for 'Explain machine learning':
- Machine Learning is a subset of AI.
- Deep Learning uses neural networks.
```